Design of Bandgap Reference Circuits Using sky130A CMOS Technology

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Introduction

The bandgap reference (BGR) circuit is one of the most important blocks in any integrated chip. Its key advantage is the remarkable stability of its output voltage under temperature variations — variations that have become increasingly significant in modern chips. As a consequence, the BGR has become an indispensable building block in analog and mixed-signal designs.

There are many topologies for realising a voltage reference that is independent of temperature. The topology adopted in this work is the **traditional BGR topology**, which consists of:

1. A **self-biased current mirror**
2. A **start-up circuit**
3. A **grid of diode-connected PNP transistors**

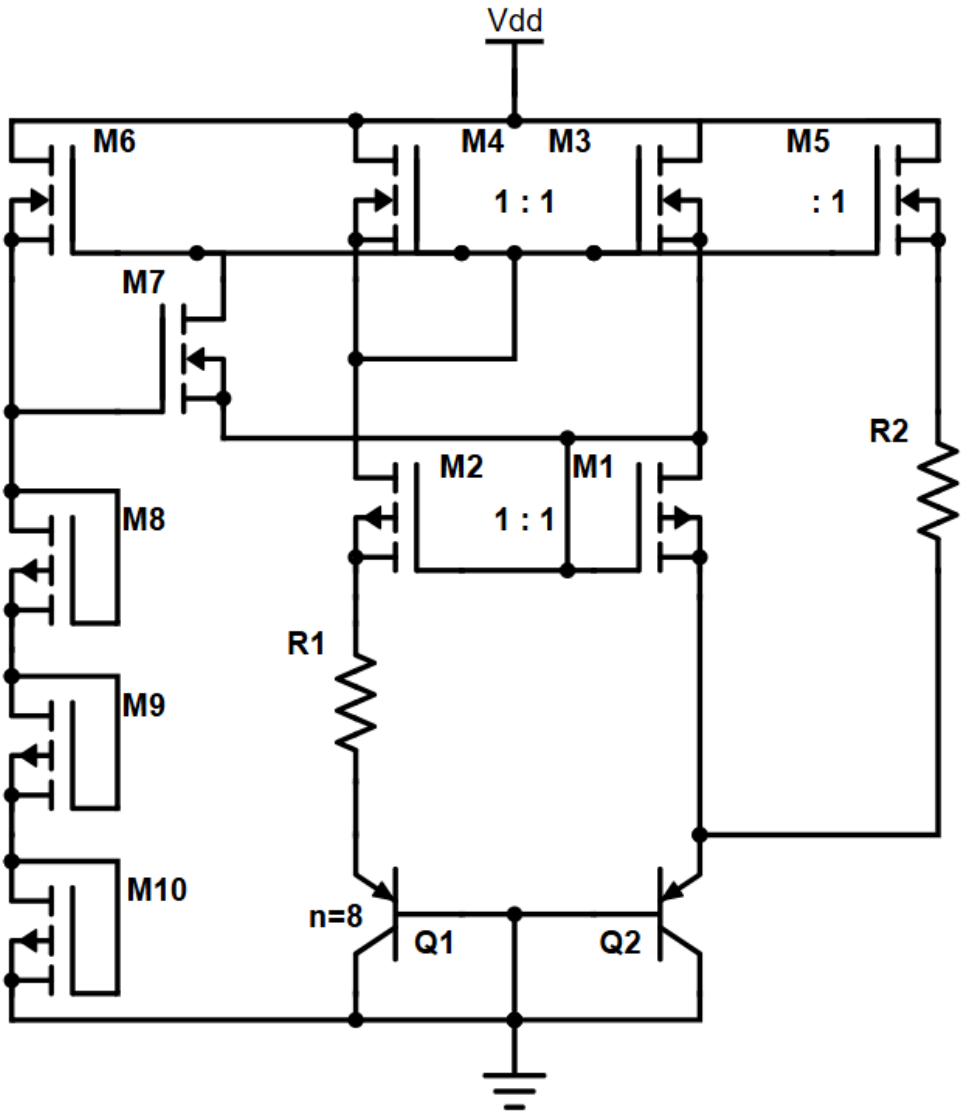


Figure 1: Traditional BGR circuit block diagram

EDA Tools and Design Flow

The entire design was carried out using **open-source EDA tools** in combination with the SkyWater **sky130A** process design kit (PDK). The three primary tools used are described below.

Tool	Role in this project
Xschem	Schematic capture and SPICE netlist generation
Ngspice	SPICE simulation (DC, AC, Monte Carlo, corners)
Magic VLSI	Physical layout, DRC, netlist extraction and LVS

2.1 Xschem — Schematic Capture

- Draw the complete transistor-level BGR schematic using sky130A native device symbols (`pfet_g5v0d10v5`, `nfet_05v0_nvt`, `pnp_05v5`, `res_xhigh_po_1p41`)
- Generate a SPICE-compatible netlist exported directly to Ngspice

2.2 Ngspice — Circuit Simulation

The following simulation types were executed:

1. **DC temperature sweep** — verifying V_{ref} stability from $-40\,°\text{C}$ to $+100\,°\text{C}$
2. **AC analysis** — computing Power Supply Rejection (PSR) vs. frequency
3. **Monte Carlo simulation** — evaluating sensitivity to process and mismatch variations
4. **Four-corners simulation** — worst-case performance across SS, TT, FF, SF, FS process corners

2.3 Magic VLSI — Physical Layout and Verification

Used for the physical implementation and sign-off of the BGR circuit:

- Full transistor-level layout drawn manually in Magic using sky130A technology rules
- **DRC** (Design Rule Check) performed interactively inside Magic — result: **0 violations**
- **Netlist extraction** performed using Magic's built-in extractor to generate a SPICE netlist from the layout
- **LVS** (Layout Versus Schematic) comparison between the Magic-extracted netlist and the Xschem schematic netlist — result: *Circuits match uniquely*

Design Procedure

3.1 Bias Current Selection

A moderate bias current is chosen:

$$I_1 = 10\,\mu\text{A}$$

The resistor R_1 controls the branch current. Its value is derived as:

$$R_1 = \frac{V_T \ln(8 \times 2)}{I_1} = 7\,k\Omega$$

where V_T is the thermal voltage:

$$V_T \approx 26\,\text{mV} \quad \text{(at room temperature)}$$

Notes on the factor of 2 in the logarithm:

- The "2" inside the ln term arises from *current doubling* through transistor Q_1.
- This doubling allows equal transistor sizing for the current-mirror devices, which improves matching and permits a lower R_1 value (hence smaller area).

3.2 Reference Voltage (V_{ref}) Derivation

V_{ref} is the sum of a **CTAT** component (the PNP base-emitter voltage, which decreases with temperature) and a **PTAT** component (the voltage across R_2, which increases with temperature):

$$V_{\text{ref}} = V_{BE} + V_{R_2} = V_{BE} + \alpha \cdot V_T$$

The coefficient α is given by:

$$\alpha = \frac{R_2}{R_1} \cdot \ln(8 \times 2) = \frac{\text{slope}_{\text{CTAT}}}{\text{slope}_{\text{PTAT}}}$$

3.3 Component Value Calculation

Knowing that :

$$\text{slope}_{\text{CTAT}} = -1.6\,\text{mV}/^\circ\text{C} \qquad \text{slope}_{\text{PTAT}} = 0.086\,\text{mV}/^\circ\text{C}$$

Then:

$$\boxed{R_2 = 46\,k\Omega \quad \text{(simulated value: } 40.8\,k\Omega\text{)}}$$

Bandgap Transistor-Level Circuit Design

The transistor-level schematic was drawn in **Xschem** using sky130A PDK primitive symbols and then exported as a SPICE netlist for simulation in **Ngspice**. The circuit uses:

- **PMOS devices:** `pfet_g5v0d10v5` (5 V, 3.3 V gate-oxide)
- **NMOS devices:** `nfet_05v0_nvt` (native threshold)
- **Resistors:** `res_xhigh_po_1p41`
- **PNP transistors:** `pnp_05v5_W3p40L3p40`

Key transistor ratios and sizing:

- Current mirror transistors M_1/M_2: ratio **1:1** (with $W/L = 25/5$)
- M_4/M_3 mirror ratio: **1:1**
- Start-up transistors M_8–M_{10}: $W/L = 1/5$
- Q_1: multiplicity $n = 8$; Q_2: multiplicity $m = 1$

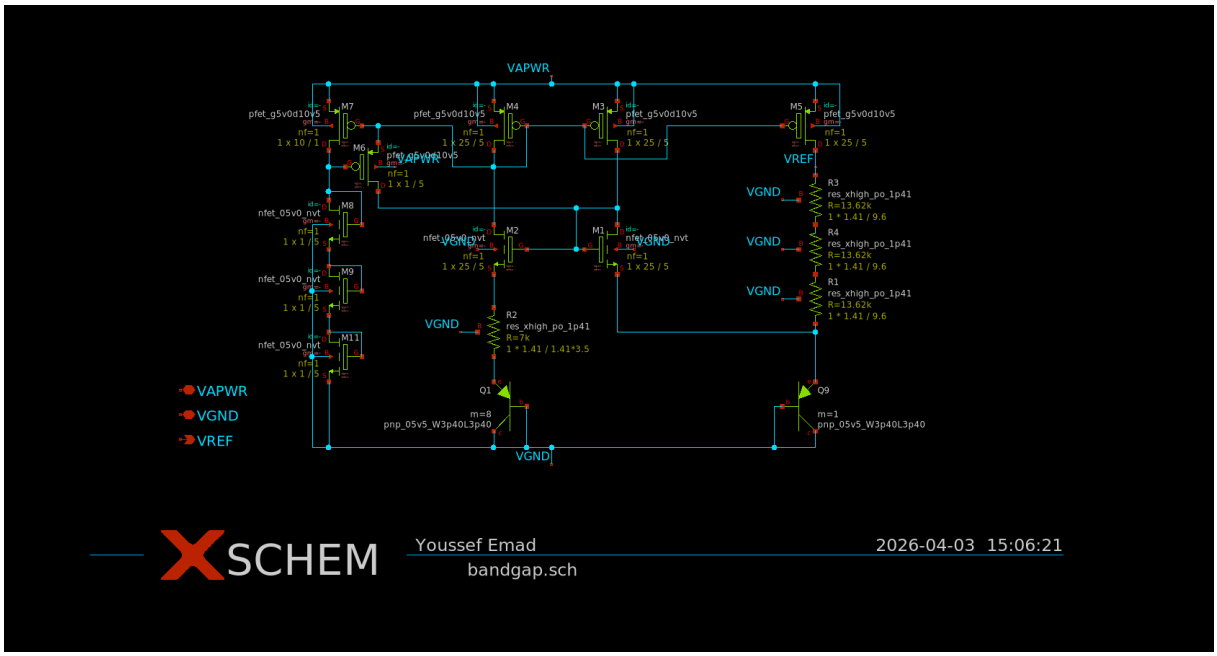


Figure 2: Transistor-level schematic of the BGR circuit in sky130A

Simulation Results

All simulations were performed using **Ngspice** with the sky130A PDK model files. The netlist was generated from the **Xschem** schematic and fed directly into Ngspice simulation decks.

5.1 Performance Summary

The following table summarises the key simulated performance metrics of the BGR circuit:

Parameter	Value
V_{ref} @ 27 °C	**1.201 V**
ΔV_{ref}	0.9 mV
Temperature Coefficient (TC)	5.35 ppm/°C
Power Supply	3.3 V
Minimum Power Supply	2.4 V
Line Regulation	18 mV
PSR @ 1 kHz	−35.4 dB

Table 1: BGR simulation performance summary

5.2 Simulation Charts

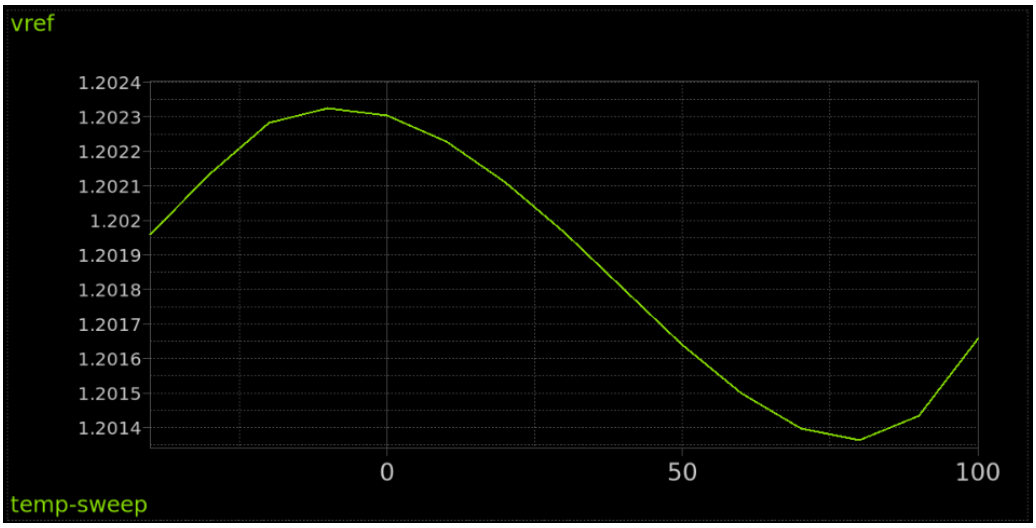


Figure 3: V_{ref} vs. temperature sweep (−40 °C to +100 °C)

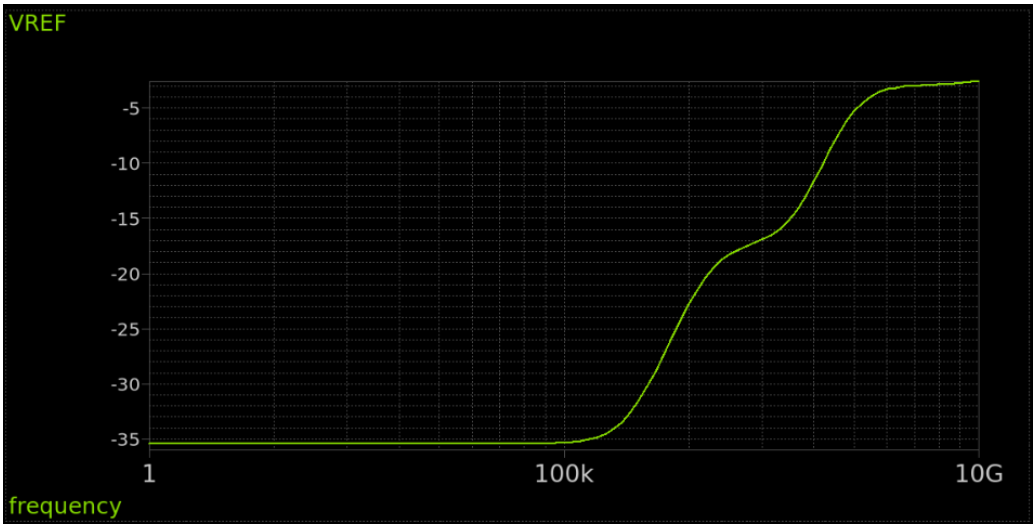


Figure 4: Power Supply Rejection (PSR) vs. frequency

Monte Carlo Simulation

Monte Carlo analysis was performed in **Ngspice** using the sky130A statistical model parameters to evaluate the circuit's sensitivity to process and mismatch variations across multiple runs. The plot below shows V_{ref} vs. temperature for all Monte Carlo samples, demonstrating the spread in the reference voltage due to manufacturing variability.

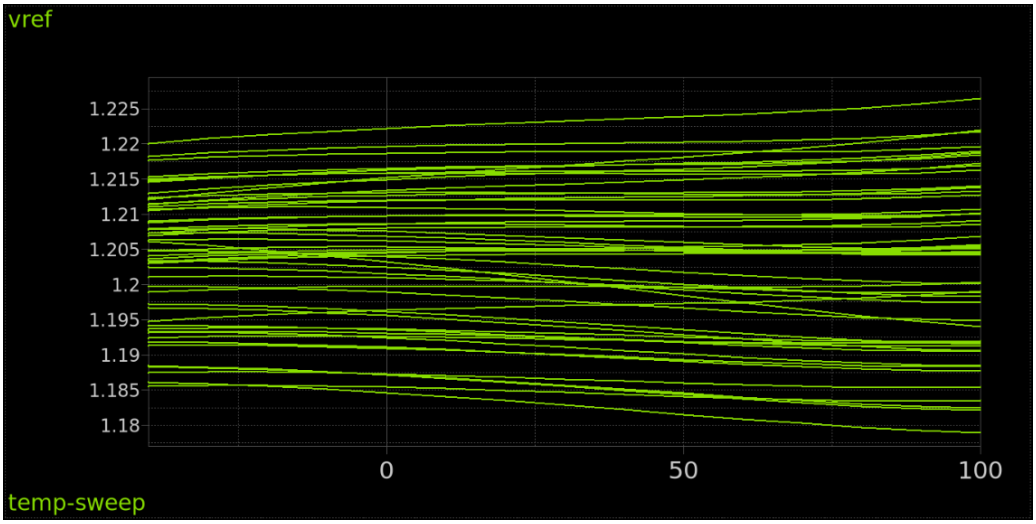


Figure 5: Monte Carlo simulation: V_{ref} vs. temperature (multiple process/mismatch runs)

The Monte Carlo results confirm that despite process spread, the BGR maintains a stable reference across the full temperature range, with V_{ref} spanning approximately 1.18 V to 1.225 V across all samples.

Four-Corners Simulation

Four-corners (worst-case process corner) simulation was performed in **Ngspice** across the SS, SF, FS, and FF corners of the sky130A PDK. The reference voltage and temperature coefficient were verified to remain within acceptable bounds across all corners, confirming the robustness of the design under worst-case manufacturing conditions.

Process corners evaluated:

- **TT** — Typical-Typical (nominal)
- **SS** — Slow-Slow (worst-case slow)
- **FF** — Fast-Fast (worst-case fast)
- **SF** — Slow NMOS / Fast PMOS
- **FS** — Fast NMOS / Slow PMOS

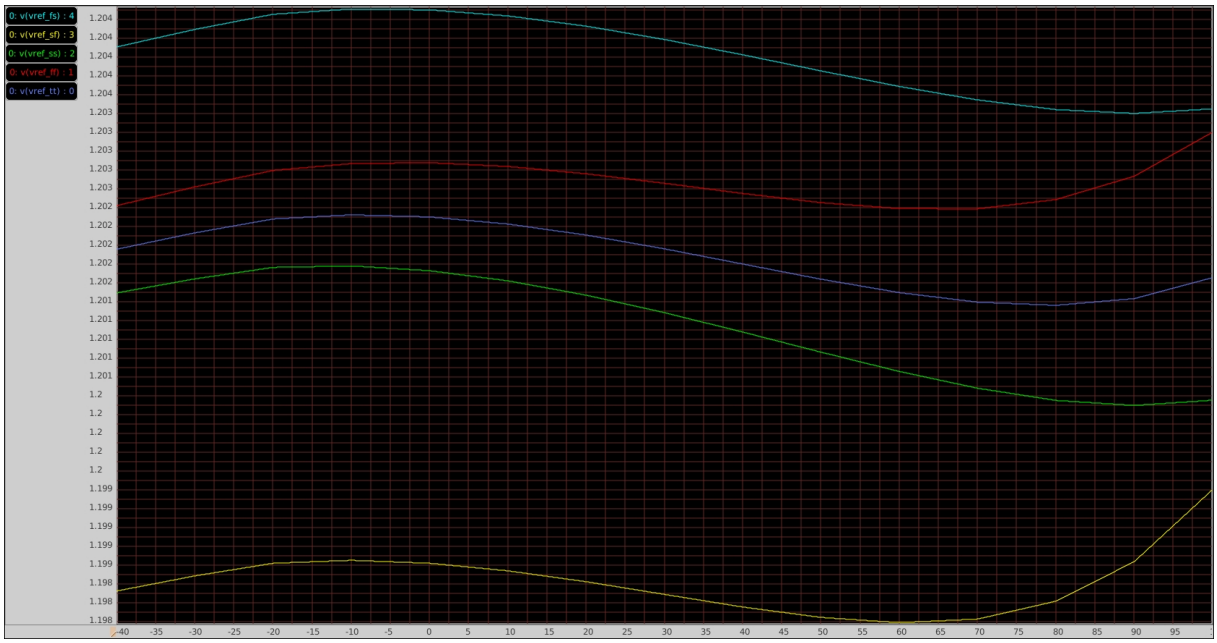


Figure 6: Four-Corners Simulation

Layout Implementation and Verification

8.1 Layout

The physical layout of the BGR circuit was implemented in the sky130A PDK using Magic VLSI. The total layout area is:

$$\boxed{60\,\mu\text{m} \times 46\,\mu\text{m}}$$

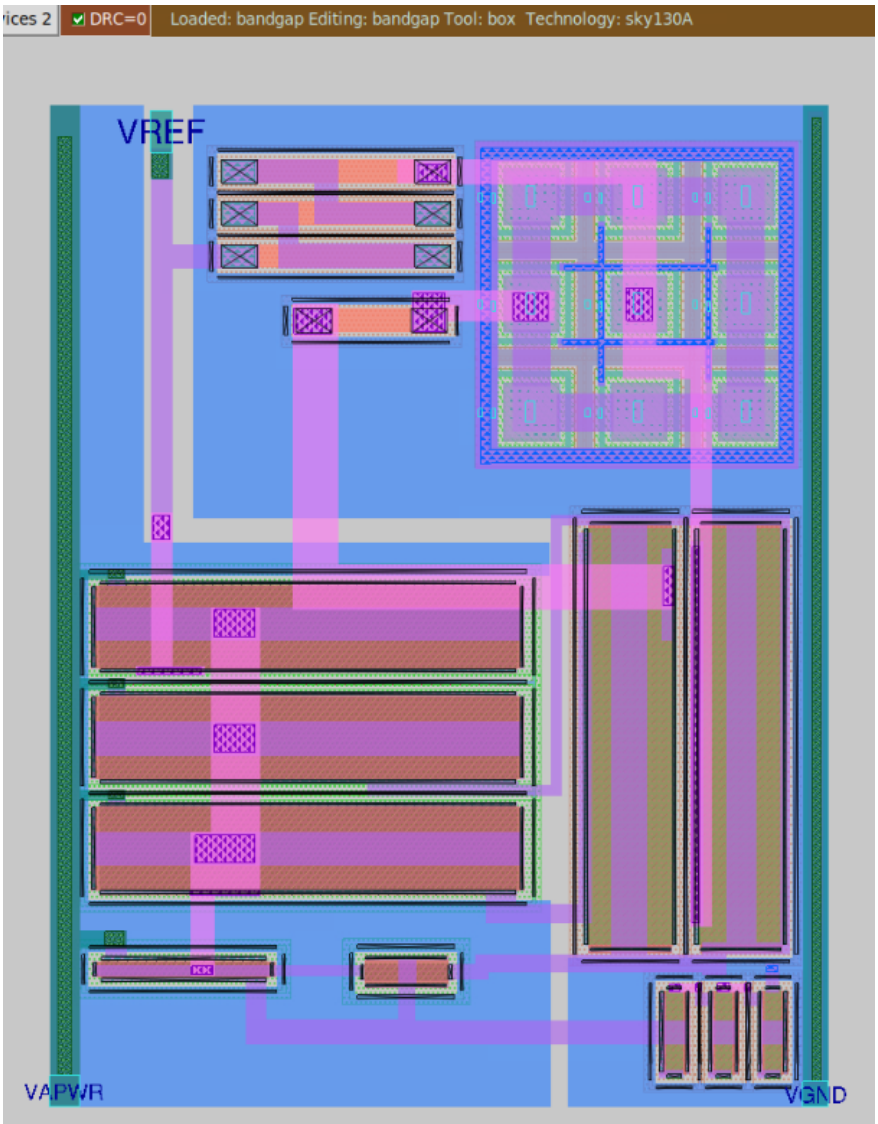


Figure 7: BGR layout in Magic VLSI (sky130A) — DRC: 0 errors

8.2 LVS Verification

Layout Versus Schematic (LVS) verification was performed by comparing the SPICE netlist extracted from **Magic** against the original **Xschem** schematic netlist. The LVS output confirms a unique and correct match:

```
Circuit 1: bandgap                          |Circuit 2: bandgap
--------------------------------------------|--------------------------------------------
sky130_fd_pr__pnp_05v5_W3p40L3p40 (9->2)    |sky130_fd_pr__pnp_05v5_W3p40L3p40 (9->2)
sky130_fd_pr__pfet_g5v0d10v5 (5)            |sky130_fd_pr__pfet_g5v0d10v5 (5)
sky130_fd_pr__nfet_05v0_nvt (5)             |sky130_fd_pr__nfet_05v0_nvt (5)
sky130_fd_pr__res_xhigh_po_1p41 (4->2)      |sky130_fd_pr__res_xhigh_po_1p41 (4->2)
Number of devices: 14                       |Number of devices: 14
Number of nets: 11                          |Number of nets: 11
-------------------------------------------------------------------------------------
Netlists match uniquely.

Subcircuit pins:
Circuit 1: bandgap                          |Circuit 2: bandgap
--------------------------------------------|--------------------------------------------
VAPWR                                       |VAPWR
VGND                                        |VGND
VREF                                        |VREF
-------------------------------------------------------------------------------------
Cell pin lists are equivalent.
Device classes bandgap and bandgap are equivalent.

Final result: Circuits match uniquely.
.
```

Figure 8: LVS verification result — “Circuits match uniquely”

Key LVS statistics:

- **Number of devices:** 14
- **Number of nets:** 11
- **Device classes:** bandgap and bandgap are equivalent
- **Final result:** *Circuits match uniquely*

Test Project on Tiny Tapeout Server

9.1 GitHub Repository

The project is hosted on GitHub and submitted to the Tiny Tapeout 10 (TT10) analog shuttle:

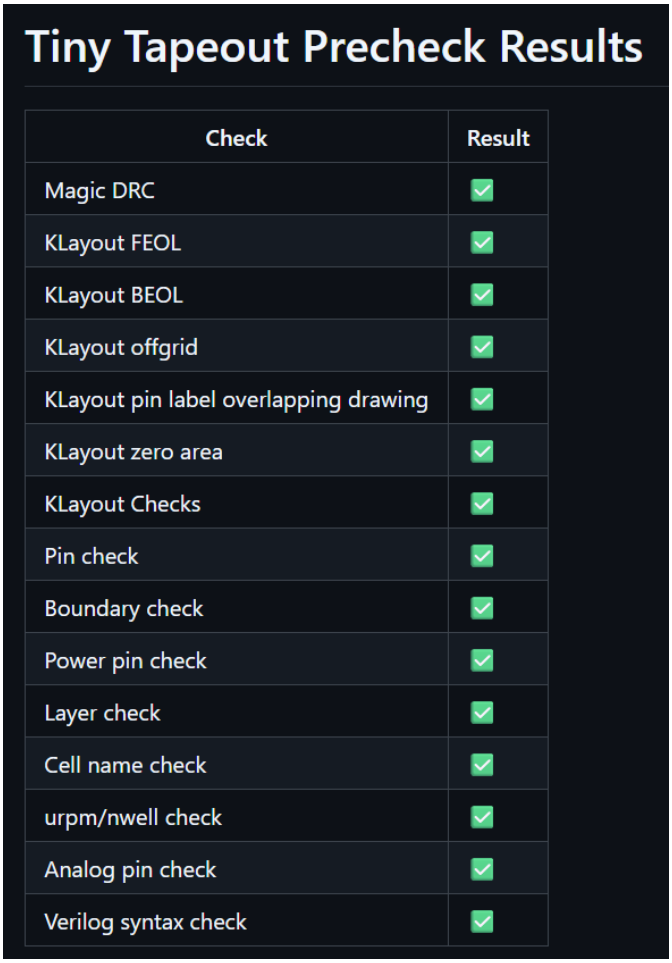
github.com/youssef0000oooo/tt10_analog_BGR

The repository achieved successful CI runs for both the **docs** and **GDS** jobs (commit `92dec93`, ver_35, pushed 2 April 2026):

Figure 9: GitHub Actions CI results — docs and GDS pipelines passing

9.2 Tiny Tapeout Precheck Results

All Tiny Tapeout precheck tests passed with no violations:



Tiny Tapeout Precheck Results

Check	Result
Magic DRC	✅
KLayout FEOL	✅
KLayout BEOL	✅
KLayout offgrid	✅
KLayout pin label overlapping drawing	✅
KLayout zero area	✅
KLayout Checks	✅
Pin check	✅
Boundary check	✅
Power pin check	✅
Layer check	✅
Cell name check	✅
urpm/nwell check	✅
Analog pin check	✅
Verilog syntax check	✅

Figure 10: Tiny Tapeout Precheck Results — all checks passed